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26 August 1977
SUPERSEDING
Fed. Std. No. 184a
23 December 1966

FEDERAL STANDARD

IDENTIFICATION MARKING OF ALUMINUM, MAGNESIUM, AND TITANIUM

Authority. This standard is issued pursuant to the Federal Property and Administrative Services Act of 1949, as amended, and its application to the purchase of commodities referred to herein is mandatory on all Federal agencies.

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TABLE I. Marking information

Subject to the limitation of column (3), markings as specified in 4.1 through 4.1.4 of this standard shall be marked on each item of aluminum, magnesium, and titanium. In addition, size in inches shall be marked on each item as specified in column (2).

(1) Item	(2) Size in inches (1) thickness (2) diameter (3) wall thickness	(3) Application
Bar, rod, straightened wire, and extruded and drawn shapes		Printed, stamped, or otherwise permanently marked on all item 1/2 inch or more in width of flat, (with less than 1/8 inch, indented surface) and 1/2 inch or more in diameter, in constantly recurring symbols at intervals not greater than 3 feet throughout the items length. Secured lifts,

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TABLE I. Marking Information (continued)

(1) Item	(2) Size in inches (1) thickness (2) diameter (3) wall thickness	(3) Application
		Secured lifts, bundles, and/or containers of sizes of items too small to mark shall contain two tags, bearing the required identification marking, attached to the metal at each end.
Ingot, pig, slab, and billet 1/		Impression stamped or otherwise permanently marked on each ingot, pig, slab, or billet.
Structural and rolled shapes		Printed, stamped, or otherwise permanently marked near end of each structural shape and rolled shape.
Coiled and spooled wire	X(2)	Marked on flange of spool or on tag attached to each coil.
Plate, sheet, and flat products (except circles) 2/	(X(1)	For plate, sheet, and flat products less than 0.012 inch thick (0 temper less than 0.020 inch thick): each piece shall be printed, stamped, or otherwise permanently marked near one end of the item. For plate, sheet and flat products 0.012 through 3/8 inches thick (0.020 through 3/8 inches thick for 0 temper) and less than 6 inches width: marked on the item in one row of constantly recurring symbols at intervals not greater than 3 feet. For plate, sheet, and flat products 0.012 through 3/8 inch in thickness, (0 temper 0.020 through 3/8 inch

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TABLE I. Marking Information (continued)

(1) Item	(2) Size in inches (1) thickness (2) diameter (3) wall thickness	(3) Application
		thickness) 6 inches through 60 inches in width, and 36 inches through 200 inches in length: Printed, stamped, or otherwise permanently marked on the item in rows of constantly recurring symbols. Symbols in each row shall recur at intervals not greater than 3 feet. Rows shall run lengthwise of the material and shall be on 6 inch centers across the width of the item. Every third row shall contain producer's name or trade-mark and the thickness of the item. Printing in other rows shall contain the commercial designation and specification data and shall be staggered. For plate, sheet, and flat products over 3/8 inch in thickness, over 60 inches in width, or over 200 inches in length: Printed, stamped, or otherwise marked on the item in one or two rows of constantly recurring symbols, repeated at intervals not greater than 3 feet. The row or rows shall run around the perimeter of the item on one surface. (Identification marking shall appear on one surface around the 4 edges of the item.) When two rows are used, one row shall consist of the commercial designation and specification data, and the second row shall consist

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TABLE I. Marking Information (continued)

(1) Item	(2) Size in inches (1) thickness (2) diameter (3) wall thickness	(3) Application
		of the producer's name or trademark and thickness. At the discretion of the producer, the requirements specified for plate, sheets, and flat products up through 3/8 inch in thickness, 6 through 60 inches in width, and 36 through 200 inches in length may be used.
Plate and sheet circles 2/	X(1)	For plate and sheet circles 24 inches and over in diameter: Marked in at least one location on each circle unless the circle was cut from continuously marked plate or sheet.
	X(1)	For plate and sheet circles less than 24 inches diameter: Marked on shipping containers or on tags attached to the shipping containers.
Tubular products except coiled tube	X(2)(3)	Printed, stamped, or otherwise permanently marked on all tubular items 1/2 inch or more in outside diameter, where the wall thickness is 0.029 inch or more in constantly recurring symbols, at intervals not greater than 3 feet throughout length of item. Secured lifts, bundles, and containers of sizes of items not covered above for continuous marking shall contain two tags, bearing the required identification

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TABLE I. Marking Information (continued)

(1) Item	(2) Size in inches (1) thickness (2) diameter (3) wall thickness	(3) Application
		markings, attached to the metal at each end.
Castings		Castings shall be marked as specified by the applicable detail specification or as specified by the procuring activity.
Coiled flat procducts[2]	X(1)	Printed in one or more rows near outside end.
Coiled tub	X(3)	Printed on tag attached to each coil or on tape used to tie coil.

[1]Specification data and lot number are not required to be marked on ingot, pig, slab, and billet products.

[2]Alclad one side flat sheet, plate, circles, and coiled sheet shall be marked on the bare side.

1. Purpose and scope. This standard establishes the physical item marking requirements for identification purposes for aluminum, magnesium, and titanium procured and issued for Government agencies. Compliance with this standard will insure positive identification of these metals. Shipment and inspection acceptance marking are not within the scope of this standard.

2. Reference doucments.

2.1 The following documents, of the issue in effect on the date of invitation for bids form a part of this standard to the extent specified herein.

Military Standards:

MIL-STD-409-Alloy Nomenclature and Temper Designation System for Magnesium Base Alloys.

MIL-STD-412-Alloy Designation System for Titanium.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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American National Standards Institute (ANST) Publication:

ANSI-H35.1-1972R Wrought Alluminium (Alloy and Temper Designation Systems for Aluminum).

(Application for copies should be addressed to the American Standards Association, 10 East 40th St., New York, N. Y., 10016).

3. Definitions and marking terms.

3.1 Definitions. Definitions of commodity forms and shapes shall be those commercially acceptable to the aluminum, magnesium, and titanium industries.

3.1.1 Item. An item is defined as each individual piece of a product.

3.1 Marking terms. Marking terms as used in this standard are defined as follows:

3.2.1 Producer's name or trademark. The producer's name or registered trademark used shall be that of the producer which performs the final processing or finishing operation prior to marketing the product.

3.2.2 Commercial designations. The commercial designation shall include a designator of composition and a designator of physical condition and quality.

(a) The composition designator is that designator approved by the respective nationally recognized industrial association or technical society such as the American Society for Testing and Materials and the Aluminum Association.

(b) The physical condition designator is that designator of temper or other physical condition approved by the respective nationally recognized industrial association or technical society.

Note. The composition and physical condition designators for wrought aluminum, titanium, and magnesium base alloy products shall be in accordance with ANSI-H35.1-1972, MIL-STD-412, and MIL-STD-409, respectively.

3.2.3 Specification data. The specification data shall include the number of the Government specification (including applicable slash sheet number when applicable) to which the metal was produced and the type, class, grade, and temper of the material, as applicable. Unless otherwise specified in the detail specification or contract or order, only the basic specification number

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(including slash sheet number when applicable) is required. The basic specification number does not include the suffix letter which indicates a revision or amendment number.

4. Marking requirements and application.

4.1 Marking information. Physical item identification marking requirement and application thereof to aluminum, magnesium, and titanium shall be in accordance with table I of this standard.

4.1.1 Manadatory marking of aluminum, magnesium, and titanium. Commercial designation specification data, and producer's name and trademark as required in table I shall be marked when applicable.

4.1.2 Titanium products. The applicable heat number shall be marked on titanium products in at least one location for each piece.

4.1.3 Magnesium products. The applicable lot number shall be marked on each piece in at least one location.

4.1.4 Aluminum products. The applicable lot number shall be marked in at least one location on each piece when specified in the aluminum material specification.

4.2 Permanency and legibility. Physical item marking on aluminum, magnesium, and titanium shall be such that it shall not rub off or be otherwise effaced by contact incident to normal handling, exposure to the elements, shipment and storage. Legibility of all markings shall be such as required for ready reaadability and the rquired permanency of identification and the height of identification and the height of characters shall be commensurate with the size of the item(s) being marked.

4.3 Deleterious effect. Physical item marking of these metals shall be accomplished in a manner which will not adversely affect the machining, forming, or fabrication of the material or produce stresses which would be deleterious to the functioning of the finished product. Impression stamping is considered detrimental and shall not be used except on ingot, pit, slab, billet, casting and cross sectional surfaces of bar, rod, and tube products.

5. Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard, supported by adequate justification, shall be sent to the Administration. This justification shall explain wherein the standard does not provide for essential needs. The request shall be sent in duplicate to the General Services Administration, Federal Supply Service, Specification Management Branch, Washington, D. C., 20406. The Administration will determine the appropriate action to be taken and will notify the agency.

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6. Conflict with referenced specifications. Where the requirements stated in this standard conflict with any requirement in a referenced specification, the requirements of the standard shall apply. Nature of conflict between the standard and the referenced specification shall be submitted in duplicate to the General Services Administration, Federal Supply Service, Specification Management Branch, Washington, D. C., 20406.

MILITARY INTEREST:

CUSTODIANS:

Navy-AS
Army-MR
Air Force-16

Review activities:

Army-MR, WC, MI
Navy-AS, OR, SH, YD
Air Force-16

User activity:

Army-ME

Preparing activity:

Air Force-16

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